

Physics 828: Homework Set No. 4

**Due date: Friday, February 10, 2011, 1:00pm
in PRB M2043 (Biao Huang's office)**

Total point value of set: 70 points

Problem 1 (10 pts.): Exercise 12.3.5 (Shankar, p. 316)

Problem 2 (10 pts.): Exercise 12.3.6 (Shankar, p. 316)

Problem 3 (20 pts.): Exercise 12.3.8 (Shankar, p. 317)

Problem 4 (10 pts.): Exercise 12.5.1 (Shankar, p. 325)

Problem 5 (10 pts.): Exercise 12.5.3 (Shankar, p. 329)

Problem 6 (10 pts.): Exercise 12.5.5, part (1) only (Shankar, p. 332)

Exercise 12.3.5 (p. 316)

The effective potential felt by the particle is

$$V_{\text{eff}}(\rho) = V(\rho) + \frac{\hbar^2 m^2}{2\mu \rho^2}$$

It creates a force

$$\begin{aligned}\vec{F}_{\text{eff}} &= -\vec{\nabla} V_{\text{eff}} = -\vec{\nabla} V(\rho) - \frac{\hbar^2 m^2}{2\mu} \left(-\frac{2}{\rho^3} \vec{e}_\rho \right) \\ &= \vec{F} + \frac{\hbar^2 m^2}{\mu} \frac{\vec{e}_\rho}{\rho^3}\end{aligned}$$

The force generated by the angular momentum points radially outward ($\sim \vec{e}_\rho$). Its magnitude is

$$\frac{L_z^2}{\mu \rho^2} \frac{1}{\rho} \text{ where } L_z = m\hbar \text{ is the angular momentum.}$$

Classically, $L_z = \mu \rho v_\perp$ where $v_\perp$ is the component of the velocity $\perp \vec{e}_\rho$.

$$\Rightarrow \frac{L_z^2}{\mu \rho^2} \frac{1}{\rho} = \frac{\mu^2 \rho^2 v_\perp^2}{\mu \rho^2} \frac{1}{\rho} = \mu \frac{v_\perp^2}{\rho}$$

Here $\frac{v_\perp^2}{\rho}$ is the centrifugal acceleration a , and the centrifugal force is μa . So the magnitude of the L_z -dependent force agrees with that of the classical centrifugal force.

Exercise 12.3.6 (p.316).

A particle of mass μ moving on a circle of radius a has kinetic energy $T = \frac{\mu}{2} v^2 = \frac{\mu}{2} \omega^2 a^2$. Expressed through $\vec{L} = \vec{r} \times \vec{p} \leftrightarrow L_z = a \cdot \mu \cdot \omega a = \mu \omega a^2$, we have $T = \frac{L_z^2}{2\mu a^2}$. There is no potential energy. Quantum

mechanically this is represented by the Hamiltonian

$$\hat{H} = \frac{\hat{L}_z^2}{2\mu a^2}$$

Its eigenvalue problem

$$\hat{H}|\Phi\rangle = E|\Phi\rangle$$

reads in position eigenbasis

$$\frac{1}{2\mu a^2} \left(-\hbar^2 \frac{\partial^2}{\partial \varphi^2} \right) \Phi(\varphi) = E \Phi(\varphi)$$

$$\Rightarrow \frac{\partial^2 \Phi(\varphi)}{\partial \varphi^2} = - \frac{2\mu a^2 E}{\hbar^2} \Phi(\varphi)$$

Solutions are $\Phi_m(\varphi) = e^{\pm im\varphi}$ ($m = 0, 1, 2, \dots$)

$$\text{with } m^2 = \frac{2\mu a^2 E_m}{\hbar^2} \Rightarrow \boxed{E_m = \frac{\hbar^2 m^2}{2\mu a^2}}$$

So the energy eigenvalues are $E_m = \frac{(L_z)_m^2}{2\mu a^2}$ with $(L_z)_m = m\hbar$

Each energy eigenvalue is two-fold degenerate:

$$E_m = E_{-m}$$

The two eigenfunctions corresponding to E_m describe clockwise and counterclockwise motion of the particle around the circle.

Exercise 12.3.8 (p. 317)

Mass μ , charge q , vector potential $\vec{A} = \frac{B}{2}(-y\vec{e}_x + x\vec{e}_y)$

$$(1) \vec{B} = \vec{\nabla} \times \vec{A} = \frac{B}{2} \begin{vmatrix} \vec{e}_x & \vec{e}_y & \vec{e}_z \\ \partial_x & \partial_y & \partial_z \\ -y & x & 0 \end{vmatrix} = \frac{B}{2} [\vec{e}_x(-\partial_z x) + \vec{e}_y(-\partial_z y) + \vec{e}_z(\partial_x x + \partial_y y)]$$

$$= B \vec{e}_z \checkmark$$

(2) For a classical particle, $\vec{F} = q \vec{v} \times \vec{B}$

$$\Rightarrow \vec{a} = \frac{q}{\mu} \frac{\vec{v}}{c} \times \vec{B} \quad \Rightarrow \vec{a} \perp \vec{B} \Rightarrow a_z = 0$$

$$\Rightarrow v_z = \text{const.}$$

Assume $v_z(0) = 0 \Rightarrow v_z(t) = 0 \quad \forall t \Rightarrow$ motion entirely in x-y plane

(If $v_z \neq 0 \Rightarrow$ spiral instead of circular orbit)

$$\frac{d^2 x}{dt^2} = \frac{q}{\mu c} \left(\frac{dy}{dt} B_z \right) \Rightarrow \ddot{x} = \frac{qB}{\mu c} v_y$$

$$\frac{d^2 y}{dt^2} = \frac{q}{\mu c} \left(-\frac{dx}{dt} B_z \right) \Rightarrow \ddot{y} = -\frac{qB}{\mu c} v_x \Rightarrow \ddot{y} = -\frac{qB}{\mu c} \dot{x} = -\underbrace{\left(\frac{qB}{\mu c} \right)^2}_{\omega_0^2} y$$

$$\Rightarrow v_y(t) = v_y(0) \cos(\omega_0 t - \delta) \quad \text{where } \boxed{\omega_0 = \frac{qB}{\mu c}}$$

$$v_x(t) = -\frac{1}{\omega_0} \dot{v}_y = +v_y(0) \sin(\omega_0 t - \delta)$$

$$\Rightarrow x(t) = -\frac{v_y(0)}{\omega_0} \cos(\omega_0 t - \delta) + c$$

$$y(t) = \frac{v_y(0)}{\omega_0} \sin(\omega_0 t - \delta) + c'$$

Boundary conditions: Assume $x(0) = -a, y(0) = 0$
 $\dot{x}(0) = 0, \dot{y}(0) = -\omega_0 a$

$$\Rightarrow \delta = 0, c = c' = 0, v_y(0) = \omega_0 a$$

$$\text{and } \boxed{\begin{matrix} x(t) = a \cos(\omega t) \\ y(t) = -a \sin(\omega t) \end{matrix}} \quad (\text{circle, clockwise motion for } q > 0)$$

(3) Consider Hamiltonian

$$\hat{H} = \frac{(\hat{P}_x + \frac{qB}{2c} \hat{Y})^2}{2\mu} + \frac{(\hat{P}_y - \frac{qB}{2c} \hat{X})^2}{2\mu}$$

$$= \frac{(\hat{\vec{P}} - \frac{q}{c} \hat{\vec{A}})^2}{2\mu}$$

Define $\hat{Q} \equiv \frac{c\hat{P}_x + \frac{qB}{2}\hat{Y}}{qB}$ $\hat{P} \equiv (\hat{P}_y - \frac{qB}{2c}\hat{X})$

$$[\hat{Q}, \hat{P}] = [\frac{c}{qB}\hat{P}_x + \frac{1}{2}\hat{Y}, \hat{P}_y - \frac{qB}{2c}\hat{X}]$$

$$= \frac{c}{qB} \underbrace{[\hat{P}_x, \hat{P}_y]}_0 + \frac{1}{2} \underbrace{[\hat{Y}, \hat{P}_y]}_{i\hbar} - \frac{1}{2} \underbrace{[\hat{P}_x, \hat{X}]}_{-i\hbar} - \frac{qB}{4c} \underbrace{[\hat{Y}, \hat{X}]}_0$$

$$= i\hbar \checkmark \Rightarrow \hat{Q} \text{ and } \hat{P} \text{ are canonical}$$

$$\Rightarrow \hat{H} = \frac{(\frac{qB}{c}\hat{Q})^2}{2\mu} + \frac{\hat{P}^2}{2\mu} = \frac{q^2 B^2}{2\mu c^2} \hat{Q}^2 + \frac{\hat{P}^2}{2\mu}$$

$$= \frac{\hat{P}^2}{2\mu} + \frac{\mu \omega_0^2}{2} \hat{Q}^2 \quad \text{harmonic oscillator (1-d)}$$

$$\Rightarrow \text{energy eigenvalues } \boxed{E_n = \hbar \omega_0 (n + \frac{1}{2})} \checkmark$$

These energy eigenstates are called Landau levels.
The example shows that in QM, energy levels get quantized even in cases where particles are free to travel to infinity (here to $z \rightarrow \pm\infty$).

$$\begin{aligned}
 (4) \quad \hat{H} &= \frac{1}{2\mu} \left(\hat{P}_x^2 + \frac{qB}{c} \underbrace{\hat{Y}\hat{P}_x}_{\text{commute!}} + \left(\frac{qB}{2c}\right)^2 \hat{Y}^2 + \hat{P}_y^2 - \frac{qB}{c} \hat{X}\hat{P}_y + \left(\frac{qB}{2c}\right)^2 \hat{X}^2 \right) \\
 &= \frac{\hat{P}_x^2 + \hat{P}_y^2}{2\mu} + \frac{\mu\omega_0^2}{8} (\hat{X}^2 + \hat{Y}^2) - \frac{\omega_0}{2} \underbrace{\hat{L}_z}_{\hat{X}\hat{P}_y - \hat{Y}\hat{P}_x} \\
 &= \hat{H}(\mu, \frac{\omega_0}{2}) - \frac{\omega_0}{2} \hat{L}_z
 \end{aligned}$$

where $\hat{H}(\mu, \frac{\omega_0}{2})$ is the 2-d isotropic harmonic oscillator with frequency $\frac{\omega_0}{2}$ for a particle of mass μ .

To show that $\hat{H}$ and $\hat{H}(\mu, \frac{\omega_0}{2})$ have a common set of eigenstates, we must show that

$$\begin{aligned}
 [\hat{H}, \hat{H}(\mu, \frac{\omega_0}{2})] &= [\hat{H}(\mu, \frac{\omega_0}{2}) + \hat{L}_z, \hat{H}(\mu, \frac{\omega_0}{2})] \\
 &= [\hat{L}_z, \hat{H}(\mu, \frac{\omega_0}{2})] = 0
 \end{aligned}$$

Compute away:

$$\begin{aligned}
 &[\hat{X}\hat{P}_y - \hat{Y}\hat{P}_x, \frac{\hat{P}_x^2 + \hat{P}_y^2}{2\mu} + \frac{\mu\omega_0^2}{8}(\hat{X}^2 + \hat{Y}^2)] \\
 &= \frac{1}{2\mu} \{ [\hat{X}\hat{P}_y, \hat{P}_x^2] - [\hat{Y}\hat{P}_x, \hat{P}_y^2] \} + \frac{\mu\omega_0^2}{8} \{ [\hat{X}\hat{P}_y, \hat{Y}^2] - [\hat{Y}\hat{P}_x, \hat{X}^2] \} \\
 &\quad \text{(all other commutators vanish)} \\
 &= \frac{1}{2\mu} \{ \underbrace{2\hat{P}_y\hat{P}_x}_{0} (i\hbar) - \underbrace{2\hat{P}_x\hat{P}_y}_{0} (i\hbar) \} + \frac{\mu\omega_0^2}{8} \{ \underbrace{2\hat{X}\hat{Y}}_{0} (-i\hbar) - \underbrace{2\hat{Y}\hat{X}}_{0} (-i\hbar) \}
 \end{aligned}$$

= 0 ✓

So $\hat{H}(\mu, \frac{\omega_0}{2})$ and $\hat{H}$ have a common set of eigenstates.

The eigenstates of $\hat{H}(\mu, \frac{\omega_0}{2})$ are (see exercise 12.3.7)

$$\hat{H}(\mu, \frac{\omega_0}{2}) |k, m\rangle = \underbrace{\frac{\hbar\omega_0}{2} (2k + |m| + 1)}_{E_{km}} |k, m\rangle$$

(12.3.7)

$\hat{L}_z |km\rangle$ is also an eigenstate of $\hat{H}(\mu, \frac{\omega_0}{2})$, with the same eigenvalue E_{km} :

$$\hat{H}(\mu, \frac{\omega_0}{2}) (\hat{L}_z |km\rangle) = \hat{L}_z \hat{H}(\mu, \frac{\omega_0}{2}) |km\rangle = E_{km} (\hat{L}_z |km\rangle)$$

$$\Rightarrow \hat{L}_z |km\rangle = \tilde{k} |km\rangle$$

But this is the eigenvalue equation for $\hat{L}_z$ for which we know the eigenvalues:

$$\tilde{k} = m \hbar$$

So the states $|km\rangle$ diagonalize both $\hat{H}(\mu, \frac{\omega_0}{2})$ and $\hat{L}_z$, and therefore also $\hat{H}$:

$$\begin{aligned} \hat{H} |km\rangle &= \left(\hat{H}(\mu, \frac{\omega_0}{2}) - \frac{\omega_0}{2} \hat{L}_z \right) |km\rangle \\ &= \left(\frac{\hbar \omega_0}{2} (2k + |m| + 1) - \frac{\omega_0}{2} \hbar m \right) |km\rangle \\ &= \hbar \omega_0 \left(k + \frac{1}{2} |m| - \frac{1}{2} m + \frac{1}{2} \right) |km\rangle \end{aligned}$$

The energy eigenvalues are

$$E = \hbar \omega_0 \left(k + \frac{1}{2} |m| - \frac{1}{2} m + \frac{1}{2} \right) = \begin{cases} \hbar \omega_0 (k + \frac{1}{2}) & m \geq 0 \\ \hbar \omega_0 (k - m + \frac{1}{2}) & m < 0 \end{cases}$$

These agree with the form $k=0, 1, 2, \dots; m \in \mathbb{Z}$

$$E_n = \hbar \omega_0 (n + \frac{1}{2}), (n = 0, 1, 2, \dots)$$

if we introduce the principal quantum number

$$n = k + \frac{1}{2} (|m| - m) = k - m \theta(-m)$$

Exercise 12.5.1 (p. 325)

$\vec{\Psi}(x, y)$ in $\mathbb{R}^2$. Consider infinitesimal rotation ϵ around z axis:

$$\psi_x(x, y) \longrightarrow \psi_x(x + \epsilon y, y - \epsilon x) - \epsilon \psi_y(x + \epsilon y, y - \epsilon x) = \psi'_x(x, y)$$

$$\psi_y(x, y) \longrightarrow \epsilon \psi_x(x + \epsilon y, y - \epsilon x) + \psi_y(x + \epsilon y, y - \epsilon x) = \psi'_y(x, y)$$

(given)

Let us Taylor expand the primed fields:

$$\psi'_x(x, y) = \psi_x(x, y) + \epsilon y \frac{\partial \psi_x}{\partial x} - \epsilon x \frac{\partial \psi_x}{\partial y} - \epsilon \psi_y(x, y) + O(\epsilon^2)$$

$$\psi'_y(x, y) = \psi_y(x, y) + \epsilon \psi_x(x, y) + \epsilon y \frac{\partial \psi_y}{\partial x} - \epsilon x \frac{\partial \psi_y}{\partial y} + O(\epsilon^2)$$

$$\Rightarrow \psi'_x(x, y) = \left(1 - \frac{i\epsilon}{\hbar} \left(x(-i\hbar \partial_y) - y(-i\hbar \partial_x) \right) \right) \psi_x - \frac{i\epsilon}{\hbar} (-i\hbar) \psi_y$$

$$\psi'_y(x, y) = \left(1 - \frac{i\epsilon}{\hbar} \left(x(-i\hbar \partial_y) - y(-i\hbar \partial_x) \right) \right) \psi_y - \frac{i\epsilon}{\hbar} (i\hbar) \psi_x$$

In matrix notation

$$\begin{pmatrix} \psi'_x \\ \psi'_y \end{pmatrix} = \left(\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} - \frac{i\epsilon}{\hbar} \begin{pmatrix} \hat{L}_z & 0 \\ 0 & \hat{L}_z \end{pmatrix} - \frac{i\epsilon}{\hbar} \underbrace{\begin{pmatrix} 0 & -i\hbar \\ i\hbar & 0 \end{pmatrix}}_{\hat{S}_z} \right) \begin{pmatrix} \psi_x \\ \psi_y \end{pmatrix}$$

$$= \left(\hat{1} - \frac{i\epsilon}{\hbar} \hat{J}_z \right) \begin{pmatrix} \psi_x \\ \psi_y \end{pmatrix}$$

We see that

$$\boxed{\hat{J}_z = \hat{L}_z + \hat{S}_z}$$

where $\hat{L}_z = \hat{L}_z^{(1)} \otimes \hat{1}^{(2)}$ and $\hat{S}_z = \hat{1}^{(1)} \otimes \hat{S}_z^{(2)}$

operates on arguments (x, y) of $\vec{\Psi}$

operates on components of $\vec{\Psi}$

Exercise 12.5.3 (p. 329)

(1) Since $\hat{J}_{\pm} |jm\rangle = \hbar \sqrt{(j \mp m)(j \pm m + 1)} |j, m \pm 1\rangle$

we have

$$\langle jm | \hat{J}_{\pm} | jm \rangle = 0$$

and hence

$$\frac{1}{2} \langle jm | \hat{J}_x \pm i \hat{J}_y | jm \rangle = 0 \Rightarrow \underbrace{\langle jm | \hat{J}_x | jm \rangle}_{=0} = \langle jm | \frac{\hat{J}_+ + \hat{J}_-}{2} | jm \rangle$$

$$\underbrace{\langle jm | \hat{J}_y | jm \rangle}_{=0} = \langle jm | \frac{\hat{J}_+ - \hat{J}_-}{2i} | jm \rangle$$

(2) $\hat{J}_x^2 = \left(\frac{\hat{J}_+ + \hat{J}_-}{2} \right)^2 = \frac{\hat{J}_+^2 + \hat{J}_-^2 + \hat{J}_+ \hat{J}_- + \hat{J}_- \hat{J}_+}{4}$

$$\Rightarrow \underbrace{\langle jm | \hat{J}_x^2 | jm \rangle}_{=0} = \frac{1}{4} (0 + 0 + \underbrace{\langle jm | \hat{J}_+ \hat{J}_- | jm \rangle}_{\hat{J}_+^2 + \hat{J}_-^2 + \hbar \hat{J}_z} + \underbrace{\langle jm | \hat{J}_- \hat{J}_+ | jm \rangle}_{\hat{J}_-^2 + \hat{J}_+^2 - \hbar \hat{J}_z})$$

$$= \frac{1}{4} 2 (j(j+1) \hbar^2 - m^2 \hbar^2) + \langle jm | \hbar \hat{J}_z | jm \rangle (1-1)$$

$$= \frac{\hbar^2}{2} (j(j+1) - m^2)$$

$$\hat{J}_y^2 = \left(\frac{\hat{J}_+ - \hat{J}_-}{2i} \right)^2 = \frac{-\hat{J}_+^2 - \hat{J}_-^2 + \hat{J}_+ \hat{J}_- + \hat{J}_- \hat{J}_+}{4}$$

$$\Rightarrow \underbrace{\langle jm | \hat{J}_y^2 | jm \rangle}_{=0} = \frac{\hbar^2}{2} (j(j+1) - m^2) \quad (\text{same non-zero terms})$$

(Follows also from $\hat{J}_y^2 = \hat{J}^2 - \hat{J}_x^2 - \hat{J}_z^2$.)

(3) $\Delta J_x = \sqrt{\langle J_x^2 \rangle} = \Delta J_y = \sqrt{\langle J_y^2 \rangle} = \hbar \sqrt{\frac{1}{2} (j(j+1) - m^2)}$

$$\Rightarrow \underbrace{\Delta J_x \cdot \Delta J_y}_{=0} = \frac{\hbar^2}{2} (j(j+1) - m^2) \quad \text{in } |jm\rangle$$

The uncertainty relation (9.2.9) states

$$(\Delta J_x)^2 (\Delta J_y)^2 \geq |\langle jm | (\hat{J}_x - \bar{J}_x) (\hat{J}_y - \bar{J}_y) | jm \rangle|^2 \quad \text{where } \bar{J}_x = \langle jm | \hat{J}_x | jm \rangle = 0 = \bar{J}_y$$

$$= |\langle jm | \hat{J}_x \hat{J}_y | jm \rangle|^2$$

$$= |\langle jm | \frac{\hat{J}_+^2 + \hat{J}_-^2 + \hat{J}_- \hat{J}_+ - \hat{J}_+ \hat{J}_-}{4i} | jm \rangle|^2$$

$$= |\langle jm | \frac{2\hbar J_z}{4i} | jm \rangle|^2 = \frac{1}{4} \hbar^2 m^2 \hbar^2$$

$$\Rightarrow \Delta J_x \cdot \Delta J_y \geq \frac{\hbar^2 m}{2}$$

$$\text{Let's check: } j(j+1) - m^2 - |m| = j(j+1) - |m|(|m|+1) \geq 0$$

$$\text{Since } j \geq |m|$$

Hence

$$\Delta J_x \cdot \Delta J_y = \frac{\hbar^2}{2} (j(j+1) - m^2) \geq \frac{\hbar^2}{2} |m| \quad \checkmark \quad \text{Eq. (9.2.9) is satisfied.}$$

(4) When $m = \pm j$, then

$$\Delta J_x \cdot \Delta J_y = \frac{\hbar^2}{2} j = \frac{\hbar^2}{2} |m|, \quad \text{so in the states } |j, \pm j\rangle$$

the inequality becomes an equality, and the uncertainty bound is saturated.

Exercise 12.5.5 (p.332)

For the $s = \frac{1}{2}$ representation $\hat{\vec{J}} = \frac{\hbar}{2} \hat{\vec{\sigma}}$ where

$$\hat{\vec{\sigma}} = (\hat{\sigma}_x, \hat{\sigma}_y, \hat{\sigma}_z) \text{ with } \hat{\sigma}_i^2 = \hat{1}$$

The Pauli matrices satisfy

$$[\hat{\sigma}_i, \hat{\sigma}_j]_+ = 2\delta_{ij} \hat{1}$$

$$\Rightarrow (\vec{\theta} \cdot \hat{\vec{\sigma}})^2 = \theta_i \hat{\sigma}_i \theta_j \hat{\sigma}_j = \theta_i \theta_j \left[\frac{1}{2} (\hat{\sigma}_i \hat{\sigma}_j + \hat{\sigma}_j \hat{\sigma}_i) \right]$$

$$= \theta_i \theta_j \delta_{ij} \hat{1} = \vec{\theta}^2 \hat{1}$$

$$\Rightarrow \underline{D^{1/2}[\vec{R}]} = e^{-\frac{i}{\hbar} \vec{\theta} \cdot \hat{\vec{J}}} = e^{-\frac{i}{2} \vec{\theta} \cdot \hat{\vec{\sigma}}}$$

$$= \sum_{n=0}^{\infty} \frac{1}{n!} \left(-\frac{i}{2}\right)^n (\vec{\theta} \cdot \hat{\vec{\sigma}})^n$$

$$= \sum_{k=0}^{\infty} \frac{1}{(2k)!} \left(-\frac{i}{2}\right)^{2k} \theta^{2k} \hat{1} + \sum_{k=0}^{\infty} \frac{1}{(2k+1)!} \left(-\frac{i}{2}\right)^{2k+1} \theta^{2k+1} \vec{\theta} \cdot \hat{\vec{\sigma}}$$

$$= \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k)!} \left(\frac{\theta}{2}\right)^{2k} \hat{1} - i \frac{\vec{\theta} \cdot \hat{\vec{\sigma}}}{|\vec{\theta}|} \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)!} \left(\frac{\theta}{2}\right)^{2k+1}$$

$$= \underline{\hat{1} \cos\left(\frac{\theta}{2}\right) - \frac{2i}{\hbar} (\vec{\theta} \cdot \hat{\vec{J}}) \sin\left(\frac{\theta}{2}\right)}$$

where $\hat{\vec{\theta}} = \frac{\vec{\theta}}{|\vec{\theta}|} = \hat{n}_{\vec{\theta}}$ = unit vector in $\vec{\theta}$ direction